Financial Dashboard n8n Workflow Setup Instructions

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Introduction

This document provides comprehensive instructions for setting up n8n workflows that power a financial dashboard displaying the top 100 cryptocurrencies, top 100 stocks, and financial news. The workflows fetch data from various free APIs, process it, and make it available for your frontend application.

The financial dashboard consists of four main n8n workflows:

1. **Cryptocurrency Data Workflow**: Fetches data for the top 100 cryptocurrencies using the CoinGecko API
2. **Stock Market Data Workflow**: Fetches data for top stocks using the Alpha Vantage API
3. **Financial News Workflow**: Fetches financial news using NewsAPI with fallback to RSS feeds
4. **Real-time Data Updates Webhook**: Provides endpoints for receiving real-time financial data updates

Prerequisites

Before setting up the workflows, ensure you have the following:

1. **n8n Installation**: A working n8n instance (version 0.214.0 or higher recommended)
2. **API Keys**: Free API keys for the following services:
3. Alpha Vantage (for stock data)
4. NewsAPI (for financial news)
5. CoinGecko (optional, works without API key but rate limits are higher with one)
6. **Basic Knowledge**: Familiarity with n8n workflow concepts and JSON

API Key Setup

Alpha Vantage API Key

1. Visit [Alpha Vantage API Key Registration]
2. Fill out the form to get your free API key
3. The free tier provides 25 requests per day (5 requests per minute)

NewsAPI Key

1. Visit [NewsAPI Registration]
2. Create a free account to get your API key
3. The free tier provides 100 requests per day

CoinGecko API Key (Optional)

1. Visit [CoinGecko API Plans]
2. Sign up for the free demo plan to get higher rate limits
3. Without an API key, you can still make 5-15 calls per minute

Workflow Installation

Method 1: Import via n8n UI

1. Open your n8n instance in a web browser
2. Navigate to the Workflows section
3. Click on the "Import from File" button
4. Select each of the workflow JSON files one by one:
5. `crypto_workflow.json`
6. `stock_workflow.json`

7. `news_workflow.json`
8. `webhook_workflow.json`
9. After importing, each workflow will appear in your workflows list

Method 2: Import via n8n CLI

If you have access to the n8n command line interface:

```
# Import cryptocurrency workflow
n8n import:workflow --input=/path/to/crypto_workflow.json

# Import stock market workflow
n8n import:workflow --input=/path/to/stock_workflow.json

# Import financial news workflow
n8n import:workflow --input=/path/to/news_workflow.json

# Import webhook workflow
n8n import:workflow --input=/path/to/webhook_workflow.json
```

Workflow Configuration

After importing the workflows, you need to configure them with your API keys and adjust settings as needed.

Setting Up Environment Variables

1. In your n8n instance, go to Settings > Variables
2. Add the following environment variables:
3. `ALPHA_VANTAGE_API_KEY` : Your Alpha Vantage API key
4. `NEWSAPI_API_KEY` : Your NewsAPI key
5. `COINGECKO_API_KEY` : Your CoinGecko API key (optional)

Cryptocurrency Workflow Configuration

1. Open the "Cryptocurrency Data - Top 100" workflow
2. Review the HTTP Request node:
3. The default endpoint is set to fetch the top 100 cryptocurrencies
4. If you have a CoinGecko API key, modify the URL to include it: `https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd&order=market_cap_desc&per_page=100&page=1&x_cg_demo_api`
5. Check the Schedule Trigger node:

6. Default is set to run every 15 minutes
7. Adjust based on your needs and API rate limits
8. Review the Write to File node:
9. Default path is `data/crypto_data_{{$now.format("YYYY-MM-DD")}}.json`
10. Ensure the directory exists or enable "Create Missing Directories"

Stock Market Workflow Configuration

1. Open the "Stock Market Data - Top Stocks" workflow
2. Check the Prepare Stock Batches node:
3. The default list includes 100 popular stocks
4. Modify the list based on your preferences
5. Review the Alpha Vantage Request node:
6. Ensure it's using your environment variable: `{{$env.ALPHA_VANTAGE_API_KEY}}`
7. Check the Wait node:
8. Default wait time is 12 seconds between API calls
9. Adjust based on your API tier limits
10. Review the Schedule Trigger node:
11. Default is set to run every hour
12. Adjust based on your needs and API rate limits

Financial News Workflow Configuration

1. Open the "Financial News" workflow
2. Review the NewsAPI Request node:
3. Ensure it's using your environment variable: `{{$env.NEWSAPI_API_KEY}}`
4. Default query is `finance OR stocks OR crypto`
5. Modify the query to focus on specific topics if needed
6. Check the Schedule Trigger node:
7. Default is set to run every 4 hours
8. Adjust based on your needs and API rate limits
9. Review the Reuters RSS Fallback node:
10. This serves as a backup when NewsAPI fails or hits rate limits
11. You can add additional RSS feeds if needed

Webhook Workflow Configuration

1. Open the "Real-time Financial Data Updates Webhook" workflow
2. Note the webhook URL:

3. This will be something like `https://your-n8n-instance.com/webhook/financial-updates-webhook`
4. You'll use this URL to send real-time updates to your dashboard
5. Review the Forward to Service node:
6. Default URL is `http://localhost:5678/webhook/{{$json["type"]}}`
7. Update this to point to your actual frontend service if needed

Testing the Workflows

Testing the Cryptocurrency Workflow

1. Open the "Cryptocurrency Data - Top 100" workflow
2. Click the "Execute Workflow" button
3. Check the execution results:
4. The HTTP Request node should return data from CoinGecko
5. The Process Data node should format the data properly
6. The Write to File node should save the data to a JSON file
7. Test the webhook endpoint:
8. Send a POST request to `https://your-n8n-instance.com/webhook/crypto-data-webhook`
9. You should receive the processed cryptocurrency data in response

Testing the Stock Market Workflow

1. Open the "Stock Market Data - Top Stocks" workflow
2. Click the "Execute Workflow" button
3. Check the execution results:
4. The workflow should create batches of stock symbols
5. The Alpha Vantage Request node should fetch data for each batch
6. The Process Stock Data node should combine and format the data
7. Test the webhook endpoint:
8. Send a POST request to `https://your-n8n-instance.com/webhook/stock-data-webhook`
9. You should receive the processed stock data in response

Testing the Financial News Workflow

1. Open the "Financial News" workflow
2. Click the "Execute Workflow" button
3. Check the execution results:

4. The NewsAPI Request node should fetch financial news
5. The Process News Data node should format the news data
6. If NewsAPI fails, the Reuters RSS Fallback should provide alternative data
7. Test the webhook endpoint:
8. Send a POST request to `https://your-n8n-instance.com/webhook/financial-news-webhook`
9. You should receive the processed news data in response

Testing the Webhook Workflow

1. Open the "Real-time Financial Data Updates Webhook" workflow
2. Test the webhook with a cryptocurrency update: `json { "type": "crypto", "symbol": "BTC", "name": "Bitcoin", "price": 50000, "change_24h": 1000, "change_percentage_24h": 2.5 }`
3. Test the webhook with a stock update: `json { "type": "stock", "symbol": "AAPL", "company_name": "Apple Inc.", "price": 150.25, "change": 2.75, "change_percent": 1.8 }`
4. Test the webhook with a news update: `json { "type": "news", "title": "Breaking Financial News", "source": "Financial Times", "url": "https://example.com/news", "description": "Important financial update" }`

Integration with Frontend

Option 1: Direct API Calls

Your frontend application can directly call the n8n webhook endpoints to fetch data:

```
// Fetch cryptocurrency data
fetch('https://your-n8n-instance.com/webhook/crypto-data-webhook', {
  method: 'POST'
})
.then(response => response.json())
.then(data => {
  // Process and display cryptocurrency data
  console.log(data);
});

// Fetch stock market data
fetch('https://your-n8n-instance.com/webhook/stock-data-webhook', {
  method: 'POST'
```

```
})
.then(response => response.json())
.then(data => {
  // Process and display stock market data
  console.log(data);
});

// Fetch financial news
fetch('https://your-n8n-instance.com/webhook/financial-news-
webhook', {
  method: 'POST'
})
.then(response => response.json())
.then(data => {
  // Process and display financial news
  console.log(data);
});
```

Option 2: Backend Integration

If you have a backend service, it can:

1. Periodically call the n8n webhook endpoints
2. Process and store the data
3. Serve the data to your frontend via your own API
4. Optionally implement WebSockets for real-time updates

Option 3: TradingView Widget Integration

For a simpler approach, you can use TradingView widgets alongside your n8n data:

```
<!-- TradingView Widget BEGIN -->
<div class="tradingview-widget-container">
  <div id="tradingview_chart"></div>
  <script type="text/javascript" src="https://
s3.tradingview.com/tv.js"></script>
  <script type="text/javascript">
  new TradingView.widget({
    "width": 980,
    "height": 610,
    "symbol": "NASDAQ:AAPL",
    "interval": "D",
    "timezone": "Etc/UTC",
    "theme": "light",
    "style": "1",
    "locale": "en",
    "toolbar_bg": "#f1f3f6",
    "enable_publishing": false,
```

```
    "allow_symbol_change": true,
    "container_id": "tradingview_chart"
  });
  </script>
</div>
<!-- TradingView Widget END -->
```

Troubleshooting

Common Issues and Solutions

API Rate Limiting

Issue: Workflows fail due to API rate limits being exceeded.

Solutions: - Increase the interval between workflow runs - Implement caching to reduce API calls - Use fallback data sources (as implemented in the news workflow) - Consider upgrading to paid API tiers for higher limits

Webhook Connection Issues

Issue: Unable to connect to webhook endpoints.

Solutions: - Ensure n8n is accessible from the internet (or your network) - Check if the webhook ID matches what's in the workflow - Verify that the correct HTTP method (POST) is being used - Check for any authentication or CORS issues

Data Processing Errors

Issue: Data is not being processed correctly.

Solutions: - Check the Function nodes for any JavaScript errors - Verify that the API responses match the expected format - Add more error handling in the Function nodes - Use the n8n debugger to inspect data at each step

Logging and Debugging

1. Enable detailed logging in n8n:
2. Set the `N8N_LOG_LEVEL` environment variable to `debug`
3. Restart n8n to apply the change
4. Use the n8n execution history:
5. Each workflow execution is recorded

6. You can view the input and output of each node
7. This helps identify where issues occur
8. Add console.log statements in Function nodes: `javascript console.log('Processing data:', items[0].json);`

Customization Options

Modifying Data Sources

Cryptocurrency Data

- To change the number of cryptocurrencies:
- Modify the `per_page` parameter in the CoinGecko API URL
- Example: `per_page=50` for top 50 cryptocurrencies
- To filter by specific cryptocurrencies:
- Use the `/simple/price` endpoint instead
- Example: `https://api.coingecko.com/api/v3/simple/price?ids=bitcoin,ethereum,ripple&vs_currencies=usd`

Stock Market Data

- To change the list of stocks:
- Modify the `topStocks` array in the "Prepare Stock Batches" Function node
- Add or remove stock symbols as needed
- To get more detailed stock data:
- Use different Alpha Vantage endpoints
- Example: `TIME_SERIES_DAILY` for historical daily data

Financial News

- To focus on specific news topics:
- Modify the query parameter in the NewsAPI URL
- Example: `q=cryptocurrency` for crypto-focused news
- To add more RSS feeds:

- Duplicate the Reuters RSS Fallback node
- Change the URL to another financial news RSS feed

Customizing Data Processing

You can modify the Function nodes to:

1. Add additional data fields
2. Implement custom calculations (e.g., moving averages)
3. Filter out unwanted data
4. Format data differently for your frontend

Adding New Workflows

Consider adding these additional workflows:

1. **Historical Data Workflow**: Fetches and stores historical price data
2. **Technical Analysis Workflow**: Calculates technical indicators
3. **Alert Workflow**: Sends notifications when prices cross thresholds
4. **Data Aggregation Workflow**: Combines data from multiple sources

References

1. n8n Documentation
2. CoinGecko API Documentation
3. Alpha Vantage API Documentation
4. NewsAPI Documentation
5. TradingView Widgets Documentation
6. Reuters RSS Feeds